

Date	29 June 2026
Team ID	SWTID-2026-7290
Project Name	Smart Lender
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, Bootstrap 5, JavaScript	Provides a responsive and user-friendly interface for applicants and bank officers to enter loan details and view prediction results across desktop and mobile devices.
2	Backend / Server-Side	Python, Flask	Flask is a lightweight web framework that integrates easily with machine learning models and efficiently handles user requests and prediction logic.
3	Database / Data Storage	SQLite (or MySQL if your project uses it)	Stores user information, loan application records, and prediction results. SQLite is simple for development, while MySQL is suitable for larger-scale deployments.
4	Cloud / Hosting / Deployment	IBM Cloud	Provides a secure and scalable environment to deploy the Flask application, making it accessible online with minimal infrastructure management.
5	Version Control & CI/CD	Git, GitHub	Enables source code management, team collaboration, version tracking, and continuous integration for efficient development.

6	Third-Party APIs / Other Tools	Scikit-learn, XGBoost, Pandas, NumPy, Joblib, Jupyter Notebook	Scikit-learn is used for data preprocessing and model evaluation, XGBoost for high-accuracy prediction, Pandas and NumPy for data manipulation, Joblib for saving the trained model, and Jupyter Notebook for model development and experimentation.
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